Information-Induced Conservatism Frontier of Distributed Stability Certificates

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Abstract

In distributed stability analysis, it is commonly believed that more available information leads to less conservative stability assessment. For example, a certificate based on complete system models is expected to be less conservative than one relying only on reduced-order summaries. However, when a stability criterion is conservative, is it because the analysis method fails to fully exploit the available information, or because the available information itself does not permit a less conservative certificate? This distinction has not been mathematically characterized: how should the value of information be defined, and how much conservatism is fundamentally caused by information compression in distributed certificate architectures?

This paper develops an information-based theoretical framework that provides a method-independent benchmark for these questions. The available information is modeled as an information structure that maps physical systems into indistinguishable system classes. For a fixed information structure, we define the best stability certificate achievable by any sound method and use it to decompose total conservatism into an irreducible information-induced part and an additional method-induced part. The framework then orders information structures, characterizes information sufficiency, and quantifies the stability value of additional information. It therefore distinguishes two different remedies for a conservative assessment: improving the certificate method with predetermined information, or enriching the information when the information-limited benchmark itself is restrictive.

The resulting perspective is task-oriented: the value of information is determined not by the number of reported quantities, but by whether the report resolves distinctions that matter to stability. The network–node examples show that spectral, scalar, or block-diagonal summaries can retain many node-level quantities while discarding stability-relevant relations such as modal alignment, local directional alignment, or cross-node coupling information. A local sensitivity law further provides a computable approximation of the information-induced conservatism.

1 Introduction

Distributed and compositional stability certificate is a recurring challenge in large-scale interconnected systems, where assembling and repeatedly analyzing a complete closed-loop model can be computationally expensive or impractical. Rather than reconstructing the entire model, distributed approaches seek local conditions whose satisfaction guarantees stability of the interconnection. Power-system stability studies provide prominent examples, including small-phase/passivity-based [1–3], small-gain-based [4], gain–phase mixed [5], and integral-quadratic-constraint-based methods [6]. The broader architectural idea is not specific to power systems: the certificate operates on selected component and interconnection descriptions rather than on the complete physical system. This compression makes scalable certificate possible, but it can also make the resulting certificates conservative. *Two fundamentally different sources of conservatism must therefore be distinguished: the analysis method may fail to exploit the information it receives, or the available information itself may be insufficient for the stability question being asked.*

Much of the recent progress has focused on making the resulting certificates less conservative. Passivity restrictions have been relaxed through lifting the scalar passivity indices to matrix-valued descriptions to retain directional and inter-channel information [7, 8]. Multiplier selection for broad classes of decentralized frequency-domain certificates has been formulated as a convex semidefinite program, enabling systematic searches for the least-conservative certificate within a prescribed family [9]. These developments address an important methodological question: *how effectively does a certificate use the information supplied to it?* They do not, however, answer the logically prior question: *how much can any sound certificate accomplish with that information in the first place?* In the extreme case, if a stable and an unstable system are indistinguishable under the available information, no improvement of the certificate algorithm alone can certify the stable one.

This distinction is operational rather than merely conceptual. When a distributed certificate is conservative, one may invest in a richer Lyapunov construction, a larger multiplier family, a tighter relaxation, or a more expensive optimization. Such effort is justified only if the existing information still permits a better sound certificate. If the best certificate compatible with the current interface is already restrictive, algorithmic refinement cannot recover the missing margin; the interface itself must be enriched. The central purpose of this paper is therefore to provide a benchmark that tells these two situations apart.

A simple example makes the issue concrete. Suppose a distributed certificate knows the individual node dynamics with the network Laplacian spectrum. Imagine a star-shaped interconnection: one hub is connected to all the other nodes. Suppose all subsystems have the same dynamics except one that with a stronger dynamic gain. Compare two cases: this high-gain subsystem is at the hub in the first case and at a leaf in the second. Both networks are the same star and therefore have exactly the same network spectrum. But the high-gain subsystem interacts with the rest of the system very differently at the hub than at a leaf. For a suitable coupling strength and a class of node dynamics, one placement can remain asymptotically stable while the other fails the Hurwitz stability test. A certificate that knows the individual node models but receives only the network eigenvalues sees the same report in both cases, so it cannot soundly distinguish the stable system from its unstable counterpart. *The missing information is not another eigenvalue; it is how the node dynamics are placed relative to the network's critical directions.* This example illustrates a possible information insufficiency and will be further clarified in Section 5.

This distinction turns conservatism into an information-sufficiency problem. Blackwell's comparison of experiments provides a classical way of ordering information structures according to whether one can be obtained from another by garbling [10]. Alternatively, the data-informativity framework asks whether available data are sufficient for a specified analysis or control task over all systems consistent with those data [11, 12]. In distributed stability analysis, however, the relevant object is not necessarily a data set, but the information structure imposed by the certificate architecture itself. A network spectrum, a scalar or matrix-valued component index, or a collection of independent local descriptions may identify physically different interconnected systems as identical from the certificate's viewpoint. *The fundamental benchmark is therefore not another sufficient stability condition, but the best stability statement that can be made by any sound method restricted to the same information.* To the best of our knowledge, this information-limited optimum and the resulting separation between information-induced and method-induced conservatism have not been characterized for distributed stability certificates in the literature.

Throughout the paper, "information" is used in this task-specific distinguishability sense, rather than in the Shannon sense or as a count of transmitted scalars. A higher-dimensional report can have little or even zero stability value if it does not separate the systems that determine the worst case. Mathematically, evaluating the best certificate over an information class resembles a robust worst-case calculation, but the object of study is different: robust analysis typically starts from a prescribed uncertainty set, whereas here the compatible set is induced by the information structure of certificates themselves. The question is therefore not only how to analyze a fixed compatible set, but how changing the information architecture refines that set and moves the fundamental certification limit.

This paper develops an information-based framework for answering this question. The framework is stated for general classes of interconnected systems and task-dependent stability margins; power-system

certificate interfaces serve as motivating examples rather than a restriction on its scope. Its contributions follow a benchmark–diagnosis–design logic.

1. **Benchmark: the best certificate permitted by a fixed information structure.** For a specified stability margin and information map, we define the largest sound certificate as the worst-case margin over all systems compatible with the reported information. This method-independent quantity gives the fundamental certification frontier before a particular Lyapunov function, multiplier, LMI, or numerical algorithm is chosen.

2. **Diagnosis: separating information-induced and method-induced conservatism.** The benchmark decomposes the conservatism of any sound method into an irreducible information-induced gap and an additional method-induced gap. We further characterize exact-margin and binary-stability sufficiency on the information-induced equivalence classes. In particular, a stable/non-stable pair producing the same report is a constructive witness that no sound method using that interface can remove the limitation.

3. **Design: identifying stability-relevant information rather than merely adding data.** We order information structures and quantify the stability value of refinements, then demonstrate the framework through network-side spectral compression, node-side passivity-index compression, and loss of cross-node quadratic information. The three cases reveal a common mechanism: modular summaries can preserve component-level quantities while discarding the stability relations among them. A local information–conservatism law finally provides a sensitivity-based approximation of the benefit obtainable by resolving such missing information.

Together, these results turn conservatism analysis into a decision problem: *whether the next improvement should target the certificate method or the information structure.*

The remainder of the paper is organized as follows. Section 2 introduces information structures and the optimal certificate under fixed information. Section 3 develops the corresponding equivalence-class, information-sufficiency, and fundamental-limit results. Section 4 explains the network–node decomposition and unifies the sources of information loss. Sections 5–7 then study network-side, node-side, and cross-node information losses, respectively. Section 8 develops a computable local approximation of the information frontier from stability sensitivity, and Section 9 concludes the paper.

Notation: The real and complex number sets are denoted by $\mathbb{R}$ and $\mathbb{C}$. For a matrix M, M^H denotes its Hermitian transpose, $\lambda(M)$ its eigenvalue, $\mathrm{spec}(M)$ its spectrum, and $\alpha(M) = \max \Re(\mathrm{spec}(M))$ the largest real part of its eigenvalues. The set of n-dimensional symmetric matrices is represented by $\mathbb{S}^n$.

2 Information Structure and Value for Stability Certificate

This section builds the concept framework required by the certificate question. We first specify the *ground-truth task margin* m, and then specify *what the distributed certificate can observe* through the information map $\mathcal{I}$. A concrete method is represented by a certificate map c acting only on that report. Soundness compares c with the ground truth and leads to the information-limited optimum $c^\star_{\mathcal{I}}$, which is the best certificate attainable without changing the information structure. The gap between m and $c^\star_{\mathcal{I}}$ is then the irreducible informational conservatism Δ_{info}. Finally, ordering and refinement of information structures quantify how this limit changes when additional information is supplied. This sequence—*task* $\rightarrow$ *information* $\rightarrow$ *sound certificate* $\rightarrow$ *information-limited optimum* $\rightarrow$ *conservatism*—is the organizing line used throughout the remainder of the paper.

2.1 Stability Evaluation Task and Information Structure

Consider a class of admissible interconnected systems $\Sigma \in \mathcal{S}$, where Σ contains the complete information about the node dynamics, network topology, parameters, and operating point relevant to the stability task.

Definition 1 (Ground-Truth Stability Margin)**.** *A specified stability-evaluation task is represented by a real-valued map*

$$\boxed{m : \mathcal{S} \to \mathbb{R}, \qquad \Sigma \mapsto m(\Sigma)}, \tag{1}$$

where $m(\Sigma) > 0$ if and only if Σ is stable. The quantity $m(\Sigma)$ is called the ground-truth stability margin *of the complete system Σ.*

The particular choice of m is task-dependent. For small-signal stability, a natural choice is the negative closed-loop spectral abscissa. Other margins, such as the distance to a prescribed $\mathcal{D}$-stability boundary or an admissible coupling margin, can be used without changing the framework below.

Definition 2 (Information Structure and Indistinguishable Class)**.** *A distributed or compositional stability analysis is assumed to observe the complete system only through an information map*

$$\boxed{\mathcal{I} : \mathcal{S} \to \mathcal{Z}, \qquad z = \mathcal{I}(\Sigma)}, \tag{2}$$

where $\mathcal{I}$ is called the information structure*. For a reported observation $z \in \mathcal{Z}$, define the corresponding indistinguishable system class (information fiber) as*

$$\mathcal{F}_{\mathcal{I}}(z) := \left\{\Sigma' \in \mathcal{S} : \mathcal{I}(\Sigma') = z\right\}. \tag{3}$$

The map $\mathcal{I}$ may collect local node reports, network or operating-state summaries, selected model coefficients, or any other quantities available to the certificate. All systems in $\mathcal{F}_{\mathcal{I}}(z)$ are indistinguishable to any method whose input is restricted to $\mathcal{I}$. Thus, $\mathcal{I}$ partitions the physical system class into equivalence classes, and stability certificate from incomplete information is fundamentally a task over these classes rather than over individual models. This view is closely related to data informativity [11]. The induced equivalence-class structure and its implications for fundamental conservatism are developed formally in Section 3.

Example 1 (Passivity-Based Stability Evaluation)**.** *Passivity provides a concrete example for the proposed methodology. In a standard passivity-based distributed assessment, each node is represented by a local passivity index, while the network is represented by a complementary description. A compositional interconnection condition then combines these local reports to certify the closed-loop system without requiring the certificate to reveal detailed node models. Let the matrix passivity index [8] of the i-th node be $\Pi_i \in \mathbb{S}^2$ and the network index be $\Pi_g \in \mathbb{S}^{2n}$. Define $\Pi = \mathrm{blkdiag}(\Pi_1, \ldots, \Pi_n)$; a typical condition then takes the form $\Pi - \Pi_g \succ 0$. Hence the passivity method operates on compressed information Π rather than on the complete system Σ.*

A legacy scalar interface further compresses each matrix report to $\pi_i = \lambda_{\min}(\Pi_i)$ [13]. The scalar report merges all nodes sharing the same weakest-direction index. Matrix-valued passivity retains precisely part of the information discarded by this scalarization. This passivity example will be reused for illustration throughout the paper.

2.2 Stability Certificates, Soundness, and Optimality

Definition 3 (Stability-Margin Certificate)**.** *For a fixed information structure $\mathcal{I}$, a particular stability-analysis method using only the reported information $z = \mathcal{I}(\Sigma)$ is represented by a map*

$$\boxed{c : \mathcal{Z} \to \mathbb{R}, \qquad z \mapsto c(z)}, \tag{4}$$

where $c(z)$ is the stability margin certified by that method, expressed in the same margin convention as m. A positive value $c(z) > 0$ certifies stability, whereas $c(z) \leq 0$ means only that the method cannot certify a positive margin from the available information.

Example 2. *The certificate c may be obtained from a spectral criterion, a passivity criterion, or any other stability-analysis method, provided that its output is expressed in the same stability-margin convention as m.*

The three basic objects introduced above have distinct roles: m evaluates the true margin from the complete system, $\mathcal{I}$ specifies what information is available to a distributed certificate, and c maps that available information to a certified margin. Equivalently, the complete and distributed evaluation paths are

$$\Sigma \xrightarrow{m} m(\Sigma), \qquad \Sigma \xrightarrow{\mathcal{I}} z \xrightarrow{c} c(z). \tag{5}$$

Definition 4 (Sound Stability-Margin Certificate)**.** *A certificate map $c : \mathcal{Z} \to \mathbb{R}$ is* sound *if*

$$c(\mathcal{I}(\Sigma)) \le m(\Sigma), \qquad \forall \Sigma \in \mathcal{S}. \tag{6}$$

A sound certificate may underestimate the true margin, but it cannot overestimate it.

Theorem 1 (Optimal Certificate under Fixed Information Structure)**.** *For a fixed information map $\mathcal{I}$, the pointwise largest sound certificate is*

$$\boxed{c^\star_{\mathcal{I}}(z) = \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma)}. \tag{7}$$

Proof. Let c be any sound certificate. For every $\Sigma \in \mathcal{F}_{\mathcal{I}}(z)$, (6) gives $c(z) \le m(\Sigma)$, and hence

$$c(z) \le \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma). \tag{8}$$

Conversely, the right-hand side of (7) does not exceed $m(\Sigma)$ for any system in the same indistinguishable class and is therefore sound. Thus it is the pointwise optimum among all sound certificates. □

When evaluated at a particular system, we use the shorthand $c^\star_{\mathcal{I}}(\Sigma) := c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))$. Theorem 1 immediately gives, for every sound certificate c,

$$c(\mathcal{I}(\Sigma)) \le c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) \le m(\Sigma). \tag{9}$$

Equation (9) is the central hierarchy of the framework: m is the oracle margin, $c^\star_{\mathcal{I}}$ is the best margin permitted by the available information, and c is the margin delivered by a particular sound method.

The significance of Theorem 1 is independent of the chosen analysis technique, such as a particular Lyapunov function, passivity condition, LMI, or numerical algorithm. *Even an arbitrarily powerful certificate cannot exceed $c^\star_{\mathcal{I}}$ without using information that further refines the current indistinguishable class.*

For a particular sound method c, its pointwise conservatism is

$$\Delta_c(\Sigma; \mathcal{I}) := m(\Sigma) - c(\mathcal{I}(\Sigma)) \ge 0. \tag{10}$$

Definition 5 (Informational Conservatism)**.** *For the true system Σ, the irreducible conservatism caused solely by the information structure $\mathcal{I}$ is*

$$\boxed{\Delta_{\text{info}}(\Sigma; \mathcal{I}) := m(\Sigma) - c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))}. \tag{11}$$

Combining (10) and (11) gives

$$\boxed{\Delta_c(\Sigma; \mathcal{I}) = \underbrace{m(\Sigma) - c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))}_{\text{information-induced}} + \underbrace{c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) - c(\mathcal{I}(\Sigma))}_{\text{method-induced}}} \tag{12}$$

where both terms on the right-hand side are nonnegative. *Thus* Δ_{info} *is the part of the conservatism that no sound method can remove without additional information, whereas the remaining term reflects how far a particular method is from the information-limited optimum.*

The hierarchy can be read as the short chain

$$\Sigma \xrightarrow{\mathcal{I}} z \xrightarrow[\text{best possible}]{c^\star_{\mathcal{I}}} c^\star_{\mathcal{I}}(z) \le m(\Sigma), \qquad c(z) \le c^\star_{\mathcal{I}}(z),$$

which separates two questions that are otherwise easy to conflate: whether the *information* is sufficient enough, and whether the *chosen method* uses that information well. Figure 1 visualizes this hierarchy together with its decisive failure witness: two systems on different sides of the stability boundary that collapse to the same report.

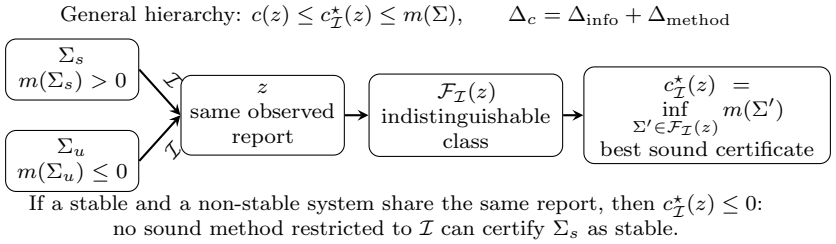


Figure 1: The information-limited benchmark and its basic failure witness. The information map makes all systems in $\mathcal{F}_{\mathcal{I}}(z)$ indistinguishable, so the best sound certificate is determined by the worst margin in that class. If the class crosses the stability boundary, no refinement of the certificate algorithm alone can certify its stable member; the information structure must be refined. The pairwise statement is formalized in Corollary 1.

2.3 Information Order and Stability Value

Definition 6 (Deterministic Informativeness Order)**.** *If there exists a deterministic map* f *such that*

$$\mathcal{I}_1 = f \circ \mathcal{I}_2, \tag{13}$$

we write $\mathcal{I}_2 \succeq \mathcal{I}_1$ *and say that* $\mathcal{I}_2$ *is at least as informative as* $\mathcal{I}_1$ *for distributed stability evaluations.*

This is the deterministic counterpart of the Blackwell garbling order [10]. Strictly, $\succeq$ is a preorder on information maps. After identifying maps that can be deterministically recovered from each other, it becomes a partial order on informational equivalence classes. Moreover, if $\mathcal{I}_2 \succeq \mathcal{I}_1$, every $\mathcal{I}_2$-fiber is contained in the corresponding $\mathcal{I}_1$-fiber. Therefore, for any $\Sigma \in \mathcal{S}$,

$$c^\star_{\mathcal{I}_2}(\mathcal{I}_2(\Sigma)) \ge c^\star_{\mathcal{I}_1}(\mathcal{I}_1(\Sigma)),\ \Delta_{\text{info}}(\Sigma;\mathcal{I}_2) \le \Delta_{\text{info}}(\Sigma;\mathcal{I}_1). \tag{14}$$

Example 3 (Scalar-to-Matrix Passivity Refinement)**.** *Returning to Example 1, the scalar summary is a deterministic function of the matrix summary through the minimum-eigenvalue map. Therefore*

$$\mathcal{I}_{\text{mat}} \succeq \mathcal{I}_{\text{sc}}, \qquad \mathcal{F}_{\text{mat}}(\Sigma) \subseteq \mathcal{F}_{\text{sc}}(\Sigma). \tag{15}$$

For small-signal stability, choose the task margin $m(\Sigma) = m_\alpha(\Sigma) := -\alpha(\Sigma)$*, where* $\alpha(\Sigma)$ *is the closed-loop spectral abscissa. Then*

$$c^\star_{\mathcal{I}_{\text{mat}}}(\mathcal{I}_{\text{mat}}(\Sigma)) \ge c^\star_{\mathcal{I}_{\text{sc}}}(\mathcal{I}_{\text{sc}}(\Sigma)). \tag{16}$$

Thus, a reduction in conservatism from scalar to matrix passivity information can be interpreted without first selecting a particular matrix LMI: the matrix report removes some adverse directional systems that remain compatible with the scalar report. Whether a specific certificate c *fully exploits this additional information is a separate question; Section 6 quantifies this information loss explicitly.*

If a refinement augments $\mathcal{I}$ by additional information $\mathcal{J}$, then $c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) \geq c^\star_{\mathcal{I}}(\Sigma)$.

Definition 7 (Stability Value of Information)**.** *For additional information $\mathcal{J}$, define its local value relative to $\mathcal{I}$ at Σ as*

$$\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) := c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - c^\star_{\mathcal{I}}(\Sigma). \tag{17}$$

The definition measures information in the same unit as the stability task. A high-dimensional report can have zero value if it removes no system that determines the fiber worst case, while one directional quantity can have substantial value if it separates the critical stable and unstable branches. *In other words, the amount of reported information need not indicate its value for a specific stability-evaluation task.*

3 Conservatism Limits from Perspective of Information Equivalence Class

3.1 Information-Induced Equivalence Classes

The information map in Definition 2 does more than reduce the dimension of the available description: it partitions the physical system class according to what the certificate can and cannot distinguish. This partition is the natural object on which the fundamental information limit acts.

Proposition 1 (Information-Induced Equivalence Relation and Quotient Space)**.** *For a fixed information structure $\mathcal{I} : \mathcal{S} \to \mathcal{Z}$, define the relation $\sim_{\mathcal{I}}$ on $\mathcal{S}$ as*

$$\Sigma_1 \sim_{\mathcal{I}} \Sigma_2 \quad \Longleftrightarrow \quad \mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2). \tag{18}$$

Then $\sim_{\mathcal{I}}$ is an equivalence relation, and the corresponding quotient space

$$\mathcal{S}/\!\sim_{\mathcal{I}} = \{[\Sigma]_{\mathcal{I}} : \Sigma \in \mathcal{S}\} \tag{19}$$

is the space of information-induced indistinguishable system classes. Each equivalence class satisfies $[\Sigma]_{\mathcal{I}} = \mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma))$.

Proof. Reflexivity follows from $\mathcal{I}(\Sigma) = \mathcal{I}(\Sigma)$. Symmetry follows because $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2)$ implies $\mathcal{I}(\Sigma_2) = \mathcal{I}(\Sigma_1)$. Transitivity follows because $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2)$ and $\mathcal{I}(\Sigma_2) = \mathcal{I}(\Sigma_3)$ imply $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_3)$. Hence $\sim_{\mathcal{I}}$ is an equivalence relation. By definition, $[\Sigma]_{\mathcal{I}}$ contains exactly the systems with the same information representation as Σ, which is $\mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma))$. □

Consequently, a method restricted to $\mathcal{I}$ operates on $\mathcal{S}/\!\sim_{\mathcal{I}}$, rather than on an individual physical realization. The stability task is therefore preserved exactly only when the information partition is sufficiently fine for that task.

3.2 Task Sufficiency and Within-Class Conservatism

By Theorem 1, the pointwise informational conservatism can be written directly on the equivalence class as

$$\boxed{\Delta_{\text{info}}(\Sigma; \mathcal{I}) = m(\Sigma) - \inf_{\Sigma' \in \mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma))} m(\Sigma')}. \tag{20}$$

Thus, once the information structure is fixed, the irreducible conservatism is completely determined by the variation of the stability task within the corresponding information equivalence class.

Theorem 2 (Equivalence-Class Width and Informational Conservatism)**.** *For any reported information value $z \in \mathcal{I}(\mathcal{S})$,*

$$\sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} \Delta_{\text{info}}(\Sigma; \mathcal{I}) = \sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) - \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma). \tag{21}$$

Consequently, the worst-case informational conservatism of the information structure is

$$\overline{\Delta}_{\text{info}}(\mathcal{I}) := \sup_{\Sigma \in \mathcal{S}} \Delta_{\text{info}}(\Sigma; \mathcal{I}) = \sup_{z \in \mathcal{I}(\mathcal{S})} \left[\sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) - \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) \right]. \tag{22}$$

Proof. For fixed z, Theorem 1 gives the same value $c^\star_{\mathcal{I}}(z) = \inf_{\Sigma' \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma')$ for every system in $\mathcal{F}_{\mathcal{I}}(z)$. Taking the supremum of $m(\Sigma) - c^\star_{\mathcal{I}}(z)$ over that class proves (21). Taking the supremum over all nonempty information classes gives (22). □

Theorem 2 shows that the relevant "size" of an information equivalence class is not its geometric diameter in the full system space, but its width along the stability task m. A large unresolved set can have negligible informational conservatism if m varies little over it, whereas a narrow class can still be costly when it crosses a steep stability-relevant direction.

The zero-width case gives the exact condition under which the stability margin itself can be evaluated on the quotient space.

Theorem 3 (Exact Certificate on the Information Quotient)**.** *Let $q_{\mathcal{I}} : \mathcal{S} \to \mathcal{S}/\!\sim_{\mathcal{I}}$ denote the natural projection $q_{\mathcal{I}}(\Sigma) = [\Sigma]_{\mathcal{I}}$. The following statements are equivalent:*

1. *the information structure introduces no irreducible conservatism, i.e., $\Delta_{\text{info}}(\Sigma; \mathcal{I}) = 0$ for all $\Sigma \in \mathcal{S}$;*
2. *the ground-truth margin is constant on every information fiber, i.e.,*

$$\Sigma_1 \sim_{\mathcal{I}} \Sigma_2 \quad \Longrightarrow \quad m(\Sigma_1) = m(\Sigma_2); \tag{23}$$

3. *there exists a unique map $\bar{m}_{\mathcal{I}} : \mathcal{S}/\!\sim_{\mathcal{I}} \to \mathbb{R}$ such that*

$$m = \bar{m}_{\mathcal{I}} \circ q_{\mathcal{I}}. \tag{24}$$

Hence exact stability-margin evaluation from the compressed information is possible if and only if the stability task itself is well defined on the information-induced quotient space.

Proof. If $\Delta_{\text{info}}(\Sigma; \mathcal{I}) = 0$ for every Σ, then (11) gives $c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) = m(\Sigma)$. Therefore, any two systems in the same information fiber have the same value of m, which proves (23). Conversely, if m is constant on every fiber, then (7) gives $c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) = m(\Sigma)$ and thus $\Delta_{\text{info}}(\Sigma; \mathcal{I}) = 0$. Finally, a function on $\mathcal{S}$ is constant on the equivalence classes of $\sim_{\mathcal{I}}$ if and only if it factors uniquely through the quotient map $q_{\mathcal{I}}$, which establishes the equivalence with (24). □

Equation (24) means that exact certificate is possible precisely when the ground-truth margin does not distinguish any two systems that the information structure has already declared indistinguishable. If this factorization fails, no refinement of the certificate map c alone can eliminate the resulting conservatism; the information structure itself must be refined.

The same quotient-space result immediately contains the binary stability problem as a special case.

Example 4 (Binary Stability as Quotient Separation)**.** *For binary stability certificate, define*

$$\begin{aligned} \mathcal{S}_{\text{s}} &:= \{\Sigma \in \mathcal{S} : m(\Sigma) > 0\}, \\ \mathcal{S}_{\text{u}} &:= \{\Sigma \in \mathcal{S} : m(\Sigma) \le 0\}. \end{aligned} \tag{25}$$

Unlike exact margin evaluation, a binary decision does not require m to be constant on each equivalence class. It only requires the information compression to preserve the stability boundary:

$$\boxed{q_{\mathcal{I}}(\mathcal{S}_{\mathrm{s}}) \cap q_{\mathcal{I}}(\mathcal{S}_{\mathrm{u}}) = \varnothing}. \tag{26}$$

Equivalently, every information equivalence class must lie entirely on one side of $m = 0$. Hence binary stability is the natural sign-only consequence of Theorem 3, rather than a separate information concept.

Corollary 1 (Indistinguishable Stable/Unstable Pair)**.** *Suppose that there exist $\Sigma_{\mathrm{s}}, \Sigma_{\mathrm{u}} \in \mathcal{S}$ such that*

$$\mathcal{I}(\Sigma_{\mathrm{s}}) = \mathcal{I}(\Sigma_{\mathrm{u}}), \qquad m(\Sigma_{\mathrm{s}}) > 0, \qquad m(\Sigma_{\mathrm{u}}) \leq 0. \tag{27}$$

Then no sound method using only $\mathcal{I}$ can certify Σ_{s} as stable.

Proof. Both systems belong to the same indistinguishable class $\mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma_{\mathrm{s}})) = \mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma_{\mathrm{u}}))$. Equation (7) gives $c^{\star}_{\mathcal{I}} \leq m(\Sigma_{\mathrm{u}}) \leq 0$ on that class. □

Corollary 1 is the basic information wall: the limitation is not that a particular distributed test fails, but that the available information does not distinguish the stable system from a physically admissible unstable one. It also converts the quotient-space separation condition above into a constructive impossibility test. Rather than characterizing the entire quotient space, it is sufficient to exhibit a single stable/non-stable pair that collapses to the same information class. Such a pair is therefore a witness that the current information structure is fundamentally insufficient, irrespective of the stability criterion used. This pairwise construction will be used in Section 5.

4 Network-Node Feedback Interconnection Structure and a Unified View of Information Loss

Distributed stability methods exploit the modular structure of an interconnected system. A common physical representation separates local node dynamics from network interconnections:

$$\begin{aligned} \text{node: } & y = G \circ u,\ G = \mathrm{blkdiag}\big(G_1, \ldots, G_n\big), \\ \text{network: } & u = -N \circ y, \end{aligned} \tag{28}$$

Here G_i is the dynamics of node i and N describes the network interconnection. This representation is widely used in practical system modeling such as power grids [3, 13]. Different port selections lead to different realizations, but the architectural principle is the same: local dynamics and network interactions are separated so that certification can be assembled modularly.

For the present paper, the crucial point is that the available information to a distributed certificate may be viewed as

$$\mathcal{I}(\Sigma) = \big\{\mathcal{I}_{\mathrm{node}},\ \mathcal{I}_{\mathrm{net}},\ \mathcal{I}_{\mathrm{joint}}\big\},$$

where the first two terms summarize the node and network sides and $\mathcal{I}_{\mathrm{joint}}$ represents any retained relation between them or across nodes. A practical interface may compress either side and may omit $\mathcal{I}_{\mathrm{joint}}$ entirely. Distributed certification is therefore not only a problem of exploiting a physical decomposition; it is also a problem of designing what that decomposition is allowed to reveal to the certificate.

The physical decomposition does not, by itself, determine what the certificate knows. The three application studies therefore vary the *reporting interface*, not the underlying idea of modular interconnection. Table 1 places the three losses side by side and makes explicit the relation that each interface fails to preserve.

The three cases expose a common structural message: *stability can depend on relations among quantities that a modular interface reports separately.* Network-side compression loses the placement of network

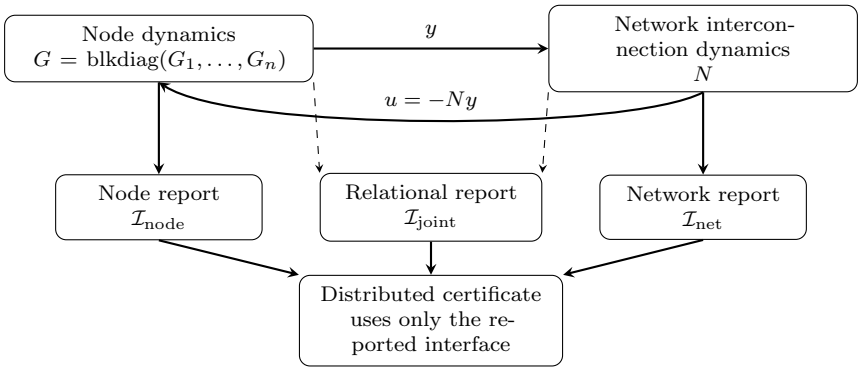


Figure 2: Physical decomposition and information-interface decomposition in distributed stability certification. The network–node model specifies how the physical system is assembled; the reporting interface specifies what the certificate can distinguish. The three studies below correspond to compression of the network report, compression of the node report, and omission of relational information, respectively.

directions relative to heterogeneous nodes; node-side compression loses local eigendirections relative to the critical interconnection direction; cross-node compression loses relations that do not belong to any single local block. In every case, the information map merges systems with potentially different stability margins into the same class. Consequently, the stability value of information is governed by how strongly the task varies along the unresolved relation, rather than by the number of scalars added to or removed from the report.

Table 1: Three representative information losses in a modular network–node architecture. The common issue is not simply that quantities are omitted, but that stability-relevant relations become unresolved.

Interface loss	What remains available	Stability-relevant relation left unresolved	Studied in
Network-side	Complete node models and the network Laplacian eigenvalues	Alignment between network eigendirections and heterogeneous node characteristics	Section 5
Node-side	Complete network certificate information and node scalar or eigenvalue-only passivity reports	Alignment between node passivity eigendirections and the critical interconnection direction	Section 6
Cross-node	Exact local matrix blocks assembled from independently certified nodes	Cross-node quadratic relations that cannot be reconstructed from the local blocks alone	Section 7

This interpretation leads to a concrete diagnostic workflow. *Step 1:* specify the stability task and its margin. *Step 2:* write the actual information map implemented by the distributed interface, including what joint relations are unavailable. *Step 3:* characterize the adverse variation inside the resulting information class, either by computing $c^\star_{\mathcal{I}}$ or by constructing an indistinguishable witness pair. *Step 4:* compare the two gaps in (12). A large method-induced gap $c^\star_{\mathcal{I}} - c$ calls for a better criterion or optimization; a large information-induced gap $m - c^\star_{\mathcal{I}}$ calls for a richer interface; if both gaps are small, the existing architecture is already close to the task-relevant limit. The following sections apply this workflow to the three interfaces in Table 1.

5 Network-Side Information Loss under Spectral Compression

The preceding results characterize the best stability certificate permitted by a given information structure. We now examine a common source of information loss on the network side of distributed assessment: replacing the network matrix by its eigenvalue spectrum. To isolate this compression effect, the node side is assumed to be known *completely*. The question is whether network eigenvalues alone, without the associated eigenvectors or their alignment with heterogeneous nodes, are sufficient for exact stability certificate.

5.1 General Heterogeneous Systems

Consider the linearized system dynamics $y = G(s)u, \qquad u = -bLy$, where $L = L^T \succeq 0$ is the Laplacian of a connected network, $b > 0$ is a strength scaling, and the complete node dynamics are

$$G(s) = g_0(s)K, \qquad K = \mathrm{diag}(k_1, \ldots, k_n) \succ 0. \tag{29}$$

Although (29) is a restricted subclass of heterogeneous nodes, an impossibility result on this subclass is already sufficient to disprove universal sufficiency of network-spectrum information for heterogeneous systems.

Since KL is similar to $K^{1/2}LK^{1/2}$, the closed-loop characteristic determinant factorizes exactly as

$$\det[I + bG(s)L] = \det[I + bg_0(s)K^{1/2}LK^{1/2}] = \prod_{j=1}^{n} \left[1 + b\mu_j g_0(s)\right], \qquad \mu_j \in \mathrm{spec}(K^{1/2}LK^{1/2}). \tag{30}$$

Thus, once the nodes are heterogeneous, the stability-relevant network quantity is generally the matched spectrum $\mathrm{spec}(K^{1/2}LK^{1/2})$, rather than $\mathrm{spec}(L)$ alone. Actually, we can derive a more impressive result as follows.

Theorem 4 (Network Spectrum Is Not Universally Sufficient under Heterogeneity)**.** *Fix any $n \geq 4$ and a stable, real-rational, strictly proper common node shape $g_0(s) = N(s)/D(s)$. Suppose that the scalar modal family*

$$\chi_\mu(s) := D(s) + \mu N(s), \qquad \mu \geq 0, \tag{31}$$

has a finite first positive-gain loss-of-Hurwitz boundary

$$0 < \mu_c := \inf\{\mu > 0 : \chi_\mu(s) \text{ is not Hurwitz}\} < \infty. \tag{32}$$

Then there exist a heterogeneous gain matrix $K \succ 0$ and two connected n-node Laplacians L_A, L_B such that

$$\mathrm{spec}(L_A) = \mathrm{spec}(L_B),$$

but one closed-loop interconnection is asymptotically stable while the other is not Hurwitz. Consequently, the information map

$$\mathcal{I}^L_{\mathrm{spec}}(\Sigma) := \{G_1(s), \ldots, G_n(s), \mathrm{spec}(L)\} \tag{33}$$

is not universally sufficient for exact stability evaluation on this heterogeneous system class. A sufficient condition for the finite-boundary assumption above is that g_0 has relative degree at least three.

Proof. In the proof, we consider the cases with star topology demonstrated in Section 1. Because D is Hurwitz, χ_μ remains Hurwitz for all sufficiently small $\mu \geq 0$, so $\mu_c > 0$. The theorem assumes that a finite loss-of-Hurwitz boundary exists. The commonly encountered relative-degree condition $r := \deg D - \deg N \geq 3$ guarantees this finiteness: the root locus of $D(s) + \mu N(s)$ has r branches tending to infinity, and, for either sign of the high-frequency gain, at least one asymptote enters the open right-half plane when $r \geq 3$. Hence χ_μ is non-Hurwitz for some finite positive μ.

It remains to construct two networks with distinct stability and the same spectrum. Let

$$K = \mathrm{diag}(\kappa, 1, \ldots, 1), \qquad \kappa > 1.$$

Consider an unweighted n-node star. In Network A, node 1, carrying the gain κ, is the center. In Network B, node 2 is the center, so the same labeled heterogeneous node 1 becomes a leaf. The two Laplacians have exactly the same complete spectrum,

$$\mathrm{spec}(L_A) = \mathrm{spec}(L_B) = \{0, 1, \ldots, 1, n\}. \tag{34}$$

The node models remain fixed at their labeled nodes and are therefore identical in the two systems.

For Network A,

$$\lambda_{L,A} := \lambda_{\max}(K^{1/2} L_A K^{1/2}) = \kappa(n-1) + 1. \tag{35}$$

For Network B, the $n-3$ directions among the ordinary leaves have eigenvalue 1, while the remaining nonzero matched eigenvalues are the roots of $p(\lambda) = \lambda^2 - (n+\kappa)\lambda + \kappa(n-1) + 1$. Hence

$$\lambda_{L,B} = \frac{n + \kappa + \sqrt{(n+\kappa)^2 - 4[\kappa(n-1)+1]}}{2}. \tag{36}$$

Substituting $\lambda_{L,A}$ gives

$$p(\lambda_{L,A}) = (\kappa - 1)(n-2)[\kappa(n-1)+1] > 0. \tag{37}$$

Moreover, for $n \geq 4$ and $\kappa > 1$, $\lambda_{L,A} > (n+\kappa)/2$, so $\lambda_{L,A}$ lies to the right of the vertex of p. Therefore

$$\lambda_{L,A} > \lambda_{L,B}. \tag{38}$$

Choose

$$b = \frac{\mu_c}{\lambda_{L,A}}.$$

For Network B, every matched eigenvalue is nonnegative and satisfies

$$b\mu_j \leq b\lambda_{L,B} = \mu_c \frac{\lambda_{L,B}}{\lambda_{L,A}} < \mu_c. \tag{39}$$

By the definition of the first boundary, every scalar factor in (30) is Hurwitz, and Network B is asymptotically stable. In Network A, the largest matched mode satisfies $b\lambda_{L,A} = \mu_c$. Since the set of Hurwitz polynomials is open and the coefficient map is continuous, the boundary polynomial χ_{μ_c} is non-Hurwitz. Hence Network A is not Hurwitz. This produces an indistinguishable stable/non-stable pair under $\mathcal{I}^L_{\mathrm{spec}}$ and proves the result. □

The theorem should be read as a statement about *universal information sufficiency*, not as a claim that every heterogeneous node model produces an isospectral stability mismatch. A single admissible heterogeneous subclass containing such a pair is enough to show that the report $\{G_i, \mathrm{spec}(L)\}$ cannot support an exact certificate that is valid for the whole class. *The missing quantity is the alignment between network modes and heterogeneous node characteristics.* This distinction disappears in the homogeneous limit $K = kI$, because then $K^{1/2} L K^{1/2} = kL$ and the matched spectrum is determined completely by $\mathrm{spec}(L)$. Thus the limitation is not caused by spectral analysis itself, but by compressing away a node–network relation that becomes stability-relevant under heterogeneity.

5.2 Quantifying the Conservatism of Spectral Compression

The preceding theorem proves that the network spectrum can be insufficient. The information framework also allows the resulting conservatism to be quantified. For the common-shape class in (29), consider the margin

$$m(\Sigma) = \mu_c - b\lambda_{\max}(K^{1/2}LK^{1/2}), \tag{40}$$

where μ_c is defined in (32).

If the full labeled network matrix L is available, this margin is known exactly. If only the spectral report (33) is available, the node matrix K remains fixed while L can vary over all admissible isospectral networks in the same information fiber. Therefore, Theorem 1 gives

$$c^{\star}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma) = \mu_c - b \sup_{\Sigma' \in \mathcal{F}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma)} \lambda_{\max}\Big(K^{1/2}L'K^{1/2}\Big). \tag{41}$$

Consequently, the exact conservatism introduced by compressing the network matrix into its spectrum is

$$\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}^L_{\mathrm{spec}}) = b\Big[\sup_{\Sigma' \in \mathcal{F}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma)} \lambda_{\max}(K^{1/2}L'K^{1/2}) - \lambda_{\max}(K^{1/2}LK^{1/2})\Big]. \tag{42}$$

Every node permutation P^TLP, while keeping the node matrix K fixed at the labeled nodes, is a physically admissible Laplacian with the same spectrum. Hence

$$\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}^L_{\mathrm{spec}}) \geq b\Big[\max_{P \text{ permutation}} \lambda_{\max}(K^{1/2}P^TLPK^{1/2}) - \lambda_{\max}(K^{1/2}LK^{1/2})\Big]. \tag{43}$$

The lower bound isolates the conservatism caused purely by the unresolved alignment of known nodes relative to a network with the same spectrum.

Example 5. *The general result above can be seen in a four-node construction for which all quantities are available in closed form. Fix the labeled node dynamics as $G_i(s) = k_i/(s+1)^3$ with $(k_1, k_2, k_3, k_4) = (\kappa, \kappa, 1, 1)$ and place them into the two networks in Fig. 3. Network A places the two high-gain nodes on adjacent nodes of the ring, whereas Network B places them on opposite nodes.*

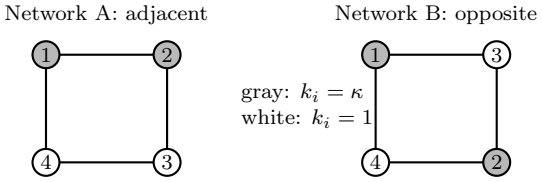


Figure 3: Four-node pair with identical labeled node dynamics and identical complete Laplacian spectrum. Only the alignment between node heterogeneity and network modes changes.

With unit line weights, the two Laplacians are

$$L_A = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}, \; L_B = \begin{bmatrix} 2 & 0 & -1 & -1 \\ 0 & 2 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{bmatrix}. \tag{44}$$

They are similar and satisfy $\mathrm{spec}(L_A) = \mathrm{spec}(L_B) = \{0, 2, 2, 4\}$. *Nevertheless, for* $K = \mathrm{diag}(\kappa, \kappa, 1, 1)$, *the largest matched eigenvalues are*

$$\lambda_{L,A} = \frac{3(\kappa+1) + \sqrt{9\kappa^2 - 14\kappa + 9}}{2}, \qquad \lambda_{L,B} = 2(\kappa+1). \tag{45}$$

Their difference is strict for every $\kappa > 1$ because $9\kappa^2 - 14\kappa + 9 - (\kappa+1)^2 = 8(\kappa-1)^2 > 0$. Take $\kappa = 2.75$ and $b = 1$. Then $\lambda_{L,A} \approx 8.72994$ and $\lambda_{L,B} = 7.5$. For $g_0(s) = 1/(s+1)^3$, the Routh–Hurwitz criterion gives the first positive boundary $\mu_c = 8$. Hence Network A is unstable while Network B is asymptotically stable.

To quantify the information loss, take Network B as the true system. Its true first-boundary margin is $m(\Sigma_B) = 8 - \lambda_{L,B} = 0.5$. The same spectral fiber contains Network A, for which $\lambda_{L,A} \approx 8.72994$. Thus

$$\Delta_{\text{info}}(\Sigma_B; \mathcal{I}^L_{\text{spec}}) \geq \lambda_{L,A} - \lambda_{L,B} \approx 1.22994, \tag{46}$$

and consequently $c^\star_{\mathcal{I}^L_{\text{spec}}}(\Sigma_B) \leq -0.72994$. Hence the spectrum-only interface loses more stability margin than the true system possesses: although Network B is stable with margin 0.5, no sound certificate based only on the common spectrum can certify it.

6 Node-Side Information Loss under Spectral Compression

The network-side result above shows that spectral compression can discard stability-relevant eigenvector alignment. We next examine the complementary node-side loss caused by compressing matrix passivity indices into scalar reports. This section uses passivity-based certificates for demonstration, but the information mechanism is not specific to passivity.

Consider the stacked passivity condition associated with Example 1,

$$\Pi - \Pi_g \succ 0, \qquad \Pi = \text{blkdiag}(\Pi_1, \ldots, \Pi_n), \tag{47}$$

where $\Pi_i \in \mathbb{S}^2$ and $\Pi_g \in \mathbb{S}^{2n}$ denote the node and network passivity indices, respectively. For this comparison, use the passivity-residual margin

$$m(\Sigma) = \lambda_{\min}(\Pi - \Pi_g) \tag{48}$$

as the task margin. The full matrix report therefore provides the benchmark against which the information loss caused by scalarization can be quantified directly.

6.1 Passivity-Based Node Reports with Scalar, Spectral, and Matrix Values

For each node, recall the scalar report $\pi_i = \lambda_{\min}(\Pi_i)$. Since $\Pi_i \in \mathbb{S}^2$, it can always be written as

$$\begin{aligned} \Pi_i &= \pi_i I_2 + r_i s_i s_i^T, \qquad \|s_i\|_2 = 1, \\ r_i &= \lambda_{\max}(\Pi_i) - \pi_i \geq 0. \end{aligned} \tag{49}$$

The full eigenvalue spectrum is $\{\pi_i, \pi_i + r_i\}$. Hence, the scalar interface retains only π_i, whereas a complete eigenvalue-spectrum report additionally reveals $\pi_i + r_i$ but still does not reveal the direction s_i. Define

$$\Pi_{\text{sc}} := \text{blkdiag}(\pi_1 I_2, \ldots, \pi_n I_2). \tag{50}$$

To avoid confusing this node-side spectrum with the network spectrum used in Section 5, denote by $\mathcal{I}^{\Pi}_{\text{spec}}$ the refinement of $\mathcal{I}_{\text{sc}}$ that reports the complete pair of eigenvalues of every Π_i, but not their eigenvectors.

Although $\mathcal{I}^{\Pi}_{\text{spec}} \succeq \mathcal{I}_{\text{sc}}$, the two information structures have exactly the same best sound certificate:

$$c^\star_{\mathcal{I}^{\Pi}_{\text{spec}}} = c^\star_{\mathcal{I}_{\text{sc}}} = \lambda_{\min}(\Pi_{\text{sc}} - \Pi_g). \tag{51}$$

To see this, every matrix report Π' compatible with either report satisfies $\Pi' \succeq \Pi_{\text{sc}}$, and therefore

$$\lambda_{\min}(\Pi' - \Pi_g) \geq \lambda_{\min}(\Pi_{\text{sc}} - \Pi_g). \tag{52}$$

For the full spectral report, let v be a unit eigenvector associated with $\lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g)$ and partition it according to the node ports as $v = [v_1^T, \ldots, v_n^T]^T$, with $v_i \in \mathbb{R}^2$. The unreported eigenvector s_i in (49) can always be chosen orthogonal to v_i. Then

$$v^T(\Pi' - \Pi_{\mathrm{sc}})v = \sum_{i=1}^{n} r_i (s_i^T v_i)^2 = 0. \tag{53}$$

Thus $v^T(\Pi' - \Pi_g)v = \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g)$. By the Rayleigh–Ritz characterization, $\lambda_{\min}(\Pi' - \Pi_g) \le v^T(\Pi' - \Pi_g)v$; together with the preceding lower bound, equality follows. Hence (51) holds even when every r_i is known exactly. Equivalently,

$$\Delta c(\mathcal{I}_{\mathrm{spec}}^{\Pi} \mid \mathcal{I}_{\mathrm{sc}}; \Sigma) = 0. \tag{54}$$

This result makes the distinction between the *amount* and the *stability value* of information explicit. Reporting the full spectrum may add many exact numbers, yet it does not reduce the worst-case conservatism unless the associated eigenvectors are also resolved.

6.2 Stability Value of Matrix-Valued Node Information

The full matrix report removes the rotational ambiguity retained by the scalar and spectral reports. Therefore,

$$c^{\star}_{\mathcal{I}_{\mathrm{mat}}} = \lambda_{\min}(\Pi - \Pi_g) = m(\Sigma), \tag{55}$$

and the exact stability value of upgrading from scalar to matrix passivity information is

$$\Delta c(\mathcal{I}_{\mathrm{mat}} \mid \mathcal{I}_{\mathrm{sc}}; \Sigma) = \lambda_{\min}(\Pi - \Pi_g) - \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g) \ge 0. \tag{56}$$

This is exactly the informational conservatism introduced by scalarizing the matrix passivity reports.

Let v_{sc} and v_{mat} be unit eigenvectors associated with the smallest eigenvalues of $\Pi_{\mathrm{sc}} - \Pi_g$ and $\Pi - \Pi_g$, respectively, and denote their i-th node blocks by v_i^{sc} and v_i^{mat}. The same variational eigenvalue reasoning gives

$$\sum_{i=1}^{n} r_i (s_i^T v_i^{\mathrm{mat}})^2 \le \Delta c(\mathcal{I}_{\mathrm{mat}} \mid \mathcal{I}_{\mathrm{sc}}; \Sigma) \le \sum_{i=1}^{n} r_i (s_i^T v_i^{\mathrm{sc}})^2. \tag{57}$$

Thus, the value of matrix passivity information is determined not by the additional eigenvalue surplus r_i alone, but by its alignment with the critical stability eigenvector.

6.3 Node–Network Direction Matching as Joint Information

The network- and node-side losses above are two views of the same missing information. On the node side, the scalar passivity report replaces Π_i by

$$\Pi_{i,\mathrm{sc}} = \pi_i I_2, \tag{58}$$

and discards the directional surplus

$$\Pi_i - \Pi_{i,\mathrm{sc}} = r_i s_i s_i^T. \tag{59}$$

The conservatism introduced by scalar passivity can therefore be expressed through the eigenvalue perturbation between the scalar and matrix certificates. Let v_{sc} be the unit eigenvector associated with the smallest eigenvalue of $\Pi_{\mathrm{sc}} - \Pi_g$, and denote its i-th node block by v_i^{sc}. By first-order eigenvalue perturbation,

$$\Delta c(\mathcal{I}_{\mathrm{mat}} \mid \mathcal{I}_{\mathrm{sc}}; \Sigma) = \lambda_{\min}(\Pi - \Pi_g) - \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g) = \sum_{i=1}^{n} r_i (s_i^T v_i^{\mathrm{sc}})^2 + O(\|\Pi - \Pi_{\mathrm{sc}}\|^2). \tag{60}$$

Therefore, the conservatism caused by scalarization is not determined by the discarded eigenvalue surplus r_i alone. It depends on how much of this surplus is aligned with the critical eigenvector of the scalar certificate.

For each node, the relevant directional information is

$$(s_i^T v_i)^2 = \|v_i\|^2 \cos^2 \theta_i, \tag{61}$$

where θ_i denotes the relative angle between the passivity eigenvector s_i and the critical network/node-interconnection eigenvector v_i. If v_i is known but s_i remains unresolved in the information class, an admissible $s_i \perp v_i$ makes (61) zero. Conversely, if s_i is known but the critical direction remains unresolved, an admissible $v_i \perp s_i$ produces the same worst case. Hence one-sided directional information has no stability value unless the relative alignment between the two sides is resolved.

This gives a common interpretation of the two preceding applications. In Section 5, $\mathrm{spec}(L)$ retains modal strengths but removes the network eigenvectors relative to heterogeneous node coordinates; the matched matrix $K^{1/2}LK^{1/2}$ restores that relation. On the node side, a scalar or eigenvalue-only passivity report retains local strengths but removes the passivity eigenvectors relative to the critical interconnection direction. The task-relevant object in both cases is therefore not an eigenvalue or eigenvector in isolation, but their *relative alignment*.

7 Cross-Node Information Loss under Block-Diagonal Compression

The preceding section studies the conservatism caused by compressing each node's coupling information into a scalar report, i.e., by neglecting *intra-node* directional information. We now move outward and ask how much additional conservatism is introduced by neglecting *cross-node* information. In particular, even if every local matrix Π_i is known exactly, the standard distributed passivity structure still restricts the node-side matrix to $\Pi = \mathrm{blkdiag}(\Pi_1, \ldots, \Pi_n)$. This section quantifies the stability value of the cross-node information discarded by this block-diagonal architecture.

An important distinction is needed first. Let $\mathcal{I}$ denote the information structure under which the node side provides strictly local information. Under independent local certificate, the local facts supporting Π_i imply the aggregate matrix Π and nothing stronger across nodes. Thus the block-diagonal structure is not an additional approximation once $\mathcal{I}$ has already been fixed: it is the mathematical consequence of the local information architecture itself. In this sense, the classical distributed passivity composition is already optimal under $\mathcal{I}$. The question below is instead the value of refining $\mathcal{I}$ by additional certified cross-node quadratic information $\mathcal{J}$, i.e., the price of distributization measured by Definition 7.

Under the local information structure, the best sound passivity certificate is

$$c^\star_{\mathcal{I}}(\Sigma) = \lambda_{\min}(\Pi - \Pi_g). \tag{62}$$

To form an information-limited benchmark for adding joint information, keep all local diagonal blocks Π_i fixed and allow the refined interface to certify a full symmetric joint matrix $\widetilde{\Pi}$ satisfying $\widetilde{\Pi}_{ii} = \Pi_i$. The best margin available under this idealized joint refinement is then

$$c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) = \sup_{\substack{\widetilde{\Pi}=\widetilde{\Pi}^T \\ \widetilde{\Pi}_{ii}=\Pi_i}} \lambda_{\min}(\widetilde{\Pi} - \Pi_g) = \min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii})\,. \tag{63}$$

Indeed, the i-th principal block of $\widetilde{\Pi} - \Pi_g$ is always $\Pi_i - (\Pi_g)_{ii}$, so its minimum eigenvalue cannot exceed the right-hand side of (63). Equality is attained by choosing $\widetilde{\Pi}_{ij} = (\Pi_g)_{ij}$ for $i \neq j$, which cancels all cross-node blocks of the network term and leaves $\widetilde{\Pi} - \Pi_g$ block diagonal.

Therefore, the exact stability value of the cross-node information is

$$\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) = c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - c^\star_{\mathcal{I}}(\Sigma) = \min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii}) - \lambda_{\min}(\Pi - \Pi_g) \geq 0. \tag{64}$$

Hence, the price of decentralization is not a separate algebraic quantity: in the terminology of this paper, it is precisely the stability value of the cross-node relational information excluded by the local information structure. Equation (64) is the maximum certificate margin that can be recovered under the joint-information refinement above.

For compactness, define the block-diagonal and off-block parts of the network index by $\Pi_{g,\mathrm{bd}} := \mathrm{blkdiag}((\Pi_g)_{11}, \dots, (\Pi_g)_{nn})$ and $\Pi_{g,\mathrm{off}} := \Pi_g - \Pi_{g,\mathrm{bd}}$. The role of network cross-node coupling is explicit. By the Weyl inequality,

$$c^\star_{\mathcal{I}}(\Sigma) \geq c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - \|\Pi_{g,\mathrm{off}}\|_2. \tag{65}$$

Equivalently,

$$\boxed{0 \leq \Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) \leq \|\Pi_{g,\mathrm{off}}\|_2}. \tag{66}$$

Thus the off-block part of the network passivity matrix gives a global upper bound on the stability value of cross-node information. If the network has no cross-node coupling in these quadratic coordinates, then $\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$ and the distributed block-diagonal restriction is lossless.

More generally, cross-node coupling can be nonzero while its information value is still zero. From (64),

$$\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0 \tag{67}$$

if and only if

$$\Pi - \Pi_g \succeq \left[\min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii})\right] I_{2n}. \tag{68}$$

This condition has a direct directional interpretation. Suppose node i attains the minimum in (63), and let x_i be a corresponding unit eigenvector of $\Pi_i - (\Pi_g)_{ii}$. Let x denote the full vector whose only nonzero block is x_i at node i. Whenever $\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$, (68) implies

$$\Pi_{g,\mathrm{off}} x = 0. \tag{69}$$

Hence nonzero cross-node coupling does not by itself imply a positive information value. If the cross-node network coupling does not act on the locally weakest directions and does not create a more adverse direction elsewhere, the local information structure already attains the joint optimum, so that $\Delta c(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$. The conservatism of distributization is therefore governed not only by the magnitude of the neglected cross-node coupling, but by whether that relational information acts along the critical local passivity directions.

8 Computable Local Approximation of Information Frontier

The preceding sections define the exact information-limited benchmark through an optimization over the full fiber $\mathcal{F}_{\mathcal{I}}(z)$. That definition is conceptually sharp, but the corresponding optimization may be expensive in a large system. This section provides the computational bridge: when the unresolved part of an information class is locally parameterized, the information-induced conservatism can be approximated directly from the sensitivity of the stability task.

Fix an information structure $\mathcal{I}$ and consider the continuous ambiguity that remains inside one of its equivalence classes. Let $\vartheta \in \mathbb{R}^n$ collect the continuous quantities not exactly resolved by the current information structure. In the applications above, such coordinates may parameterize unresolved network–node alignment, rotations of local passivity eigendirections, or missing cross-node quadratic terms. The radius $\rho_{\mathcal{I}}$ below measures the local resolution of these hidden coordinates. Locally, write the fiber as

$$\vartheta + \mathcal{U}_{\mathcal{I}}, \qquad 0 \in \mathcal{U}_{\mathcal{I}}, \tag{70}$$

and define its radius

$$\rho_{\mathcal{I}} := \sup_{\delta \in \mathcal{U}_{\mathcal{I}}} \|\delta\|_2. \tag{71}$$

Definition 8 (Unresolved Stability Information)**.** *If $m(\vartheta)$ is differentiable at the nominal point, define*

$$\mathcal{S}_{\mathcal{I}}(\vartheta; m) = \sup_{\delta \in \mathcal{U}_{\mathcal{I}}} -\delta^T \nabla m(\vartheta). \tag{72}$$

When the task and nominal point are clear, it is abbreviated as $\mathcal{S}_{\mathcal{I}}$. The quantity $\mathcal{S}_{\mathcal{I}}$ has the same unit as m. It is the largest first-order stability-margin drop hidden by variations that the current information cannot distinguish.

Example 6. *A box-like fiber $|\delta_i| \le r_i$ yields*

$$\mathcal{S}_{\mathcal{I}} = \sum_{i=1}^{n} r_i \left| \frac{\partial m}{\partial \vartheta_i} \right|. \tag{73}$$

Thus raw uncertainty widths become comparable only after they are weighted by their relevance to the stability boundary.

Theorem 5 (Local Information–Conservatism Law)**.** *Assume that m is twice continuously differentiable on $\vartheta + \mathcal{U}_{\mathcal{I}}$ and that*

$$\|\nabla^2 m(\vartheta + \delta)\|_2 \le L_m, \qquad \forall \delta \in \mathcal{U}_{\mathcal{I}}. \tag{74}$$

Then

$$\mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2 \le \Delta_{\text{info}}(\vartheta; \mathcal{I}) \le \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{75}$$

In particular, as the information resolution increases such that $\rho_{\mathcal{I}} \to 0$,

$$\boxed{\Delta_{\text{info}} = \mathcal{S}_{\mathcal{I}} + O(\rho_{\mathcal{I}}^2)}. \tag{76}$$

Proof. Let $g = \nabla m(\vartheta)$. Taylor's theorem and (74) give, for every $\delta \in \mathcal{U}_{\mathcal{I}}$,

$$m(\vartheta) + g^T\delta - \frac{L_m}{2}\|\delta\|_2^2 \le m(\vartheta + \delta) \le m(\vartheta) + g^T\delta + \frac{L_m}{2}\|\delta\|_2^2. \tag{77}$$

Taking the infimum over δ and using

$$\inf_{\delta \in \mathcal{U}_{\mathcal{I}}} g^T\delta = -\mathcal{S}_{\mathcal{I}} \tag{78}$$

yields (75) after substitution into (20). □

Equivalently, the best certificate itself satisfies

$$m(\vartheta) - \mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2 \le c_{\mathcal{I}}^\star \le m(\vartheta) - \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{79}$$

Define the two endpoints as

$$\underline{c}_{\mathcal{I}} := m(\vartheta) - \mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2,\ \bar{c}_{\mathcal{I}} := m(\vartheta) - \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{80}$$

They provide an operational three-way decision. If $\underline{c}_{\mathcal{I}} > 0$, the current information is already sufficient to establish a positive sound margin. *If $\bar{c}_{\mathcal{I}} \le 0$, even the best sound certificate $c_{\mathcal{I}}^\star$ cannot be positive, so no alternative algorithm using the same information can certify stability.* If the interval $[\underline{c}_{\mathcal{I}}, \bar{c}_{\mathcal{I}}]$ crosses zero, the local bound is inconclusive and one should refine the information or tighten the local bound.

This relation turns the abstract benchmark into an operational local test. It estimates how much margin is hidden by the unresolved coordinates and therefore supports the same diagnosis developed above: whether the present interface is already adequate for the task or whether resolving a particular stability-sensitive direction can materially improve certification.

9 Conclusion

This paper develops an information-based framework for asking a question that precedes the choice of a particular distributed stability criterion: *what is the best sound certificate that the current information structure permits at all?* The answer provides a method-independent benchmark. Comparing a concrete certificate with this benchmark then separates information-induced conservatism from method-induced conservatism, while the associated equivalence classes characterize when the stability evaluation is fundamentally possible.

The network–node studies show why this distinction matters in modular systems. The relevant loss is often not a shortage of numerical quantities but a loss of *relations*: the alignment between network modes and heterogeneous node dynamics, or relations across independently reported nodes. These examples therefore support a common principle: information should be valued by the stability distinctions it preserves, not by its raw dimension. The local information–conservatism law provides a computable approximation of this value when the full fiber-wise benchmark is difficult to evaluate.

The resulting framework has a direct design implication. If a method lies well below the information-limited benchmark, the available information is being underused, and the certificate method should be improved. If the benchmark itself lies well below the true margin, further optimization under the same interface cannot remove the limitation and the reporting architecture must be enriched. In this sense, the framework provides a principled basis for deciding where additional complexity in distributed stability certification is actually worthwhile.

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